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Initial Backtest Report: Bollinger Bands

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1. Background and Strategy Description

The Bollinger Bands trading strategy is a simple one that is based around mean reversion. It was invented in the 1980's by John Bollinger. The strategy has three parameters, EntryZscore, ExitZscore, and rollingWindow. The strategy dictates that you enter a position if the price deviates by more than EntryZscore standard deviations from the mean. If the price crosses ExitZscore standard deviations from the mean, then you exit the position. The mean and standard deviations are computed based on the rollingWindow. At any one time the strategy has zero or one units invested and can go both long and short.

It is important to note that this is a common strategy that uses Bollinger Bands as a signal. The bands can be used in many different ways and were invented to help traders determine whether a stock was trading high or low relative to its average. Their inventor, John Bollinger even stated "There is absolutely nothing about a tag of a band that in and of itself is a signal."

2. Strategy Description and Publisher Information

2.1 Publisher Information and Backtest

I discovered this strategy in a book entitled ‘Algorithmic Trading’ by Ernest P Chan. Chan tests out the strategy on the GLD-USO spread, stating that some traders believe that when oil prices go up, so do gold prices. The logic behind this being that increases in oil prices drive up inflation, and gold is highly correlated with inflation. Chan does state that after performing statistical tests on GLD and USO, there isn’t evidence to suggest that the two are cointegrated, however notes that there might be enough short term mean reversion to profit on. To better capture this mean reversion, Chan constructs a portfolio with GLD and USO and uses linear regression to dynamically re-calculate the optimal hedge ratio each day. The value of the portfolio is graphed just below, clearly displaying something that resembles a mean reverting pattern. He then runs the Bollinger bands strategy on this portfolio, as it displays far more mean reverting behavior.

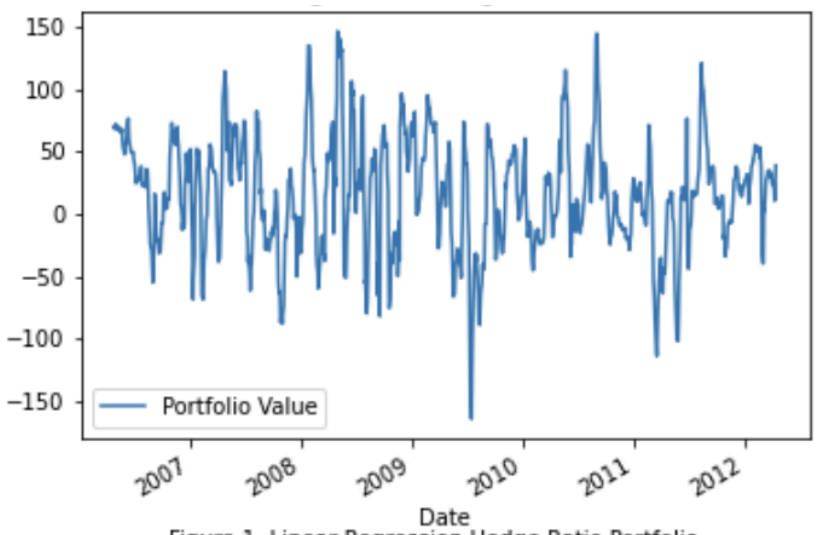

Figure 1. Linear Regression Hedge Ratio Portfolio

To capitalize on short term mean reversion, Chan set his EntryZscore to 1, ExitZscore to 0, and rollingWindow to 20, noting that when conducting a thorough backtest, these values should be optimized. Setting a short rollingWindow and a low EntryZscore means that Chan was trying to have a shorter holding period and more trades. He ran the strategy on one minute data

from May 24th 2006 until April 9th 2012. It generated an APR of 17.8% and a sharpe ratio of 0.96. The equity curve is below.

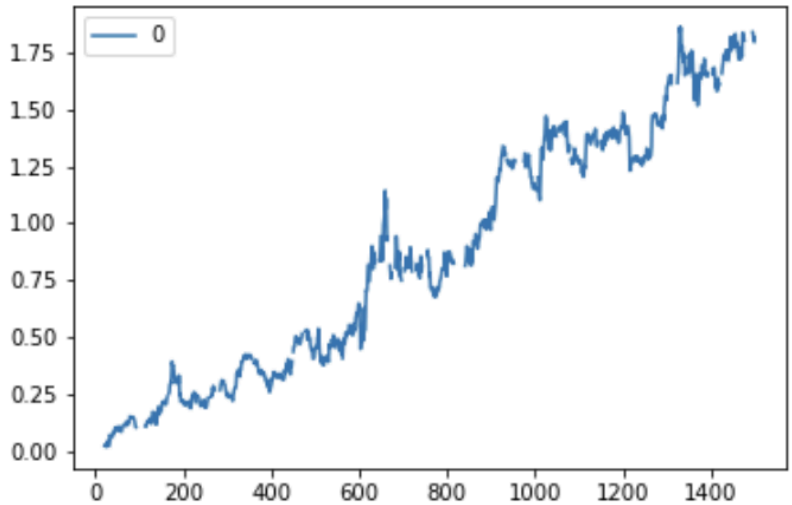

Figure 2. Cumulative Returns of Bollinger Band Strategy on GLD-USO

2.2 Replication

I attempt to reproduce the results obtained by Chan, however with some changes. Chan only used open data, while I had access to the full candle data from the time period. Chan also doesn't have any commission costs in his backtest, for which I put in a cost of 0.002. The results of the backtest are displayed below.

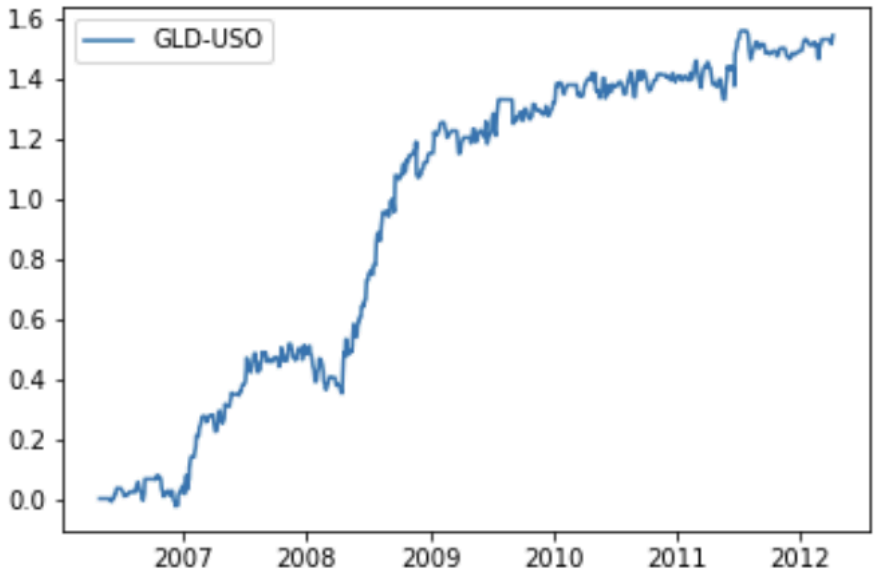

Figure 3. Returns of Bollinger Band Strategy

	GLD-USO
Strategy Return (%)	154.337
Buy and Hold Return (%)	-43.96
Sharpe Ratio	1.607
# Trades	106
Avg. Trade Duration	16 days
Expectancy (%)	133.35

Figure 4. GLD-USO Backtest Metrics

I was able to achieve a return of 154%, comparable to the ~180% achieved by Chan in his backtest. While Chan doesn't specify the exact makeup of his strategy's portfolio, my strategy traded 10 shares at a time of the hypothetical spread portfolio and started with an initial capital of $10,000. It is also important to note that since this is trading a portfolio constructed with a hedge ratio, it is important that fractional share trading be possible, otherwise the portfolio traded is purely theoretical.

3. Backtest

To test the performance of the Bollinger Bands trading strategy in other market environments, I turned to the FX markets. I go through the full backtesting procedure in this

section using 1 day EUR-GBP candles from Yahoo Finance as an example. The procedure is then repeated for twenty other currencies using both hourly and daily candles with results shown in the next section.

3.1 Assumptions

As will be shown in the following section, backtesting results differ greatly depending on the commission assumptions. Therefore, I will operate under two different assumptions. The first being a 0.2% commission fee as was done in the replication section, and the second a 0.02% commission fee, meant not to account for commissions, but rather to assume a spread cost. Additionally, all strategies tested from here on out assume that the maximum position possible is taken when a signal is received unless otherwise stated.

3.2 Default Parameters

For the first backtest, I implemented the same conditions as Chan's backtest, setting the entry and exit z-scores to 1 and 0 respectively, and the rollingWindow to 20. As mentioned, this was done in both a commission and commission free environment. The performance statistics and equity curve are below:

Figure 5. Backtest Returns of EURGBP

Figure 6. Backtest Returns of EURGBP w/ Commissions

	EURGBP	EURGBP w/ Commissions
Strategy Return (%)	53.35	-66.35
Buy and Hold Return (%)	20.27	20.27
Sharpe Ratio	0.23	0
# Trades	842	842
Avg. Trade Duration	9 days	9 days
Expectancy (%)	0.06	-0.12

Figure 7. EURGBP Basic Backtest Metrics

The results from the initial test are completely different depending on the environment. Recall from section 2 that Chan selected these parameters to have a low entry requirement and

make a lot of trades. We can see that this is the case here, specifically paying notice to the high number of trades and low expectancy in the no-commission environment. Thus it makes sense that when commissions are in place, we would lose money on more of our trades given the low expectancy of each trade. This suggests that if we are trading with high commissions, it would make sense to increase the values of our parameters in order to target larger deviations from the mean that would be profitable to trade.

3.3 Optimization

To conduct the optimization, I first split my dates to train on the first 70% of dates and test on the last 30%. I defined 3 hyperparameters: entryZscore, exitZscore, and lookback period, and then optimized for final equity value with the restraint that entryZscore must be greater than or equal to exitZscore.

The results of the optimization for each currency are shown in the charts below. The heatmaps displayed are outputted by the optimization on the training set and can be interpreted from left to right as follows.

1. Averaged over all lookback periods, which entryZ and exitZ gave the best returns?
2. Averaged over all exitZ, which entryZ and lookback periods gave the best returns?
3. Averaged over all entryZ, which exit Z and lookback periods gave the best returns?

The z-score axis in the plots are the z-score multiplied by 10, while the n axis is the lookback period in days. Based on the heatmaps we then run selected parameters on the testing data to obtain the detailed performance statistics.

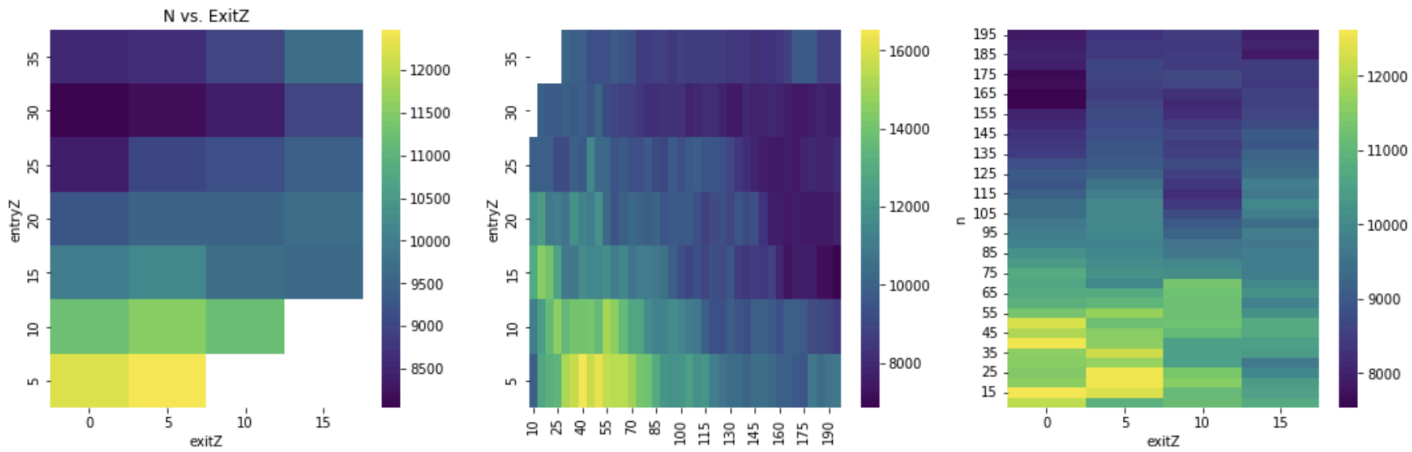

Figure 8. EURGBP Optimization Heatmaps

From the first heatmap, it is evident that we prefer values for entryZ and exitZ between 0 and 0.5. The subsequent two heatmaps are a little bit harder to interpret. It is clear that we prefer a longer lookback period, something between 40-50, and it is likely that choosing an entryZ that is slightly higher than our exitZ for this currency will have strong results.

Next, I will choose a set of near optimal parameters to run a backtest to produce statistics for the optimized strategy run on the testing data. For the test data I will choose entryZ=0.5, exitZ=0, and n=40, decided from the information provided by the optimization heatmaps. The equity curve and relevant statistics are displayed below.

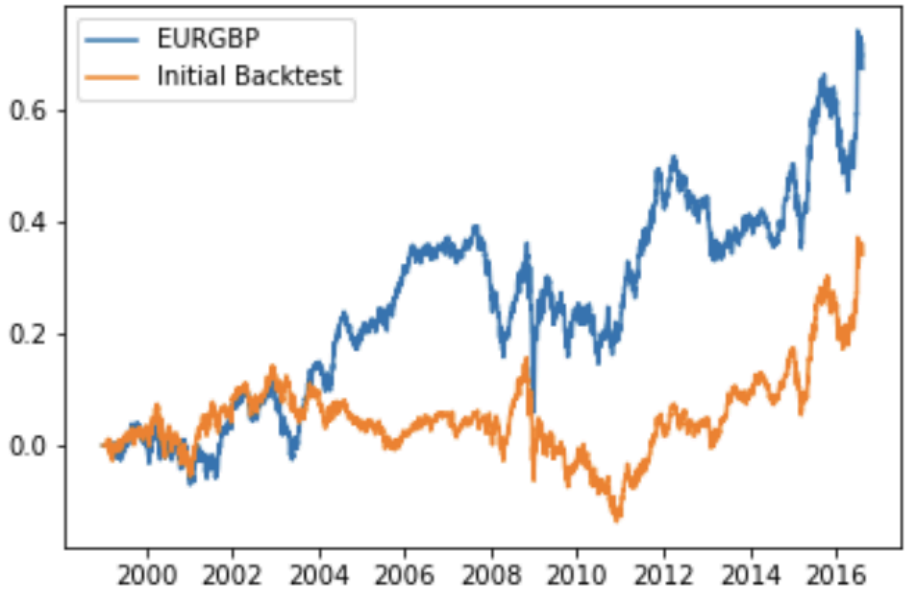

Figure 9. Optimized Backtest for EURGBP Training Data

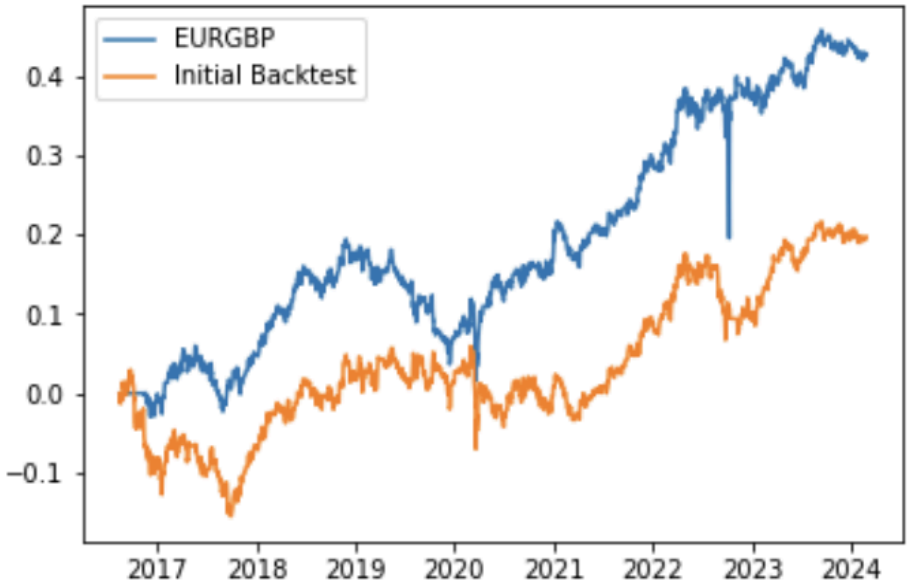

Figure 10. Optimized Backtest for EURGBP Testing Data

	EURGBP Train	EURGBP Test
Strategy Return (%)	67.2	42.75
Buy and Hold Return (%)	20.82	-0.52
Sharpe Ratio	0.34	0.49
# Trades	410	179
Avg. Trade Duration	14 days	14 days
Expectancy	0.13	0.20

Figure 11. Optimized EURGBP Backtest Metrics

We see that the optimized strategy performs well on the test data for the non-commission case. especially compared to the buy and hold return and base strategy return. The strategy generates over 43% return in a barely positive market.

Next we plot the heatmaps for the commission data to the below. It is easy to see that as earlier hypothesized, we prefer higher values for our exit and entry Z-scores, as smaller mean reversion opportunities no longer become profitable. We seem to prefer entryZscores of 3-3.5 and exitZscores of 1-1.5. Surprisingly, we still seem to prefer values of n around 50. The parameters I chose to select for the test dataset are entryZ=3, exitZ=1.5, and n=50. Those results and equity curve are displayed on the next page.

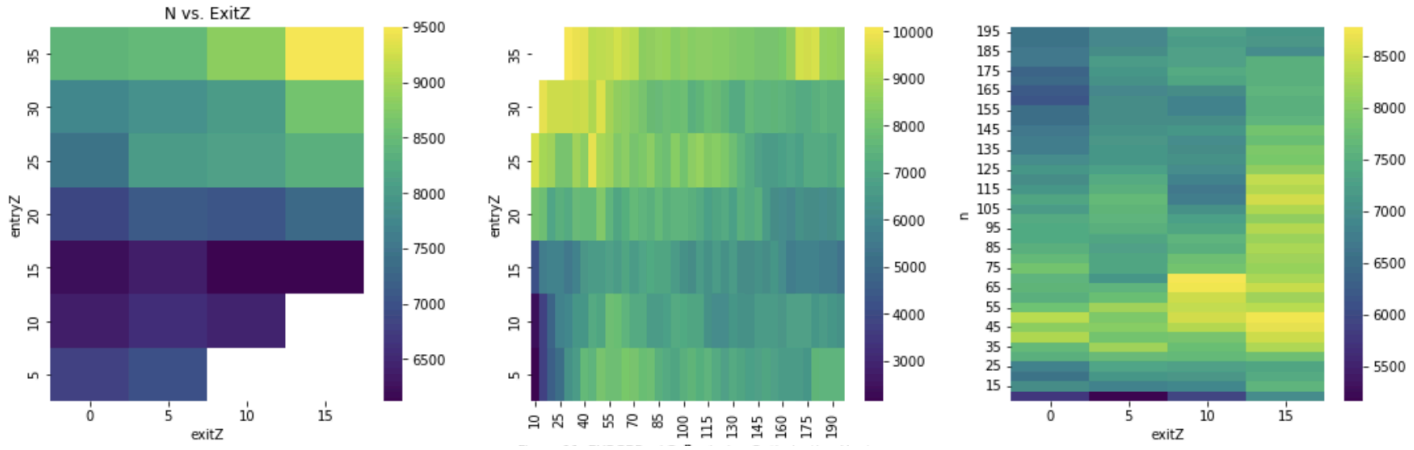

Figure 12. EURGBP w/ Commission Optimization Heatmaps

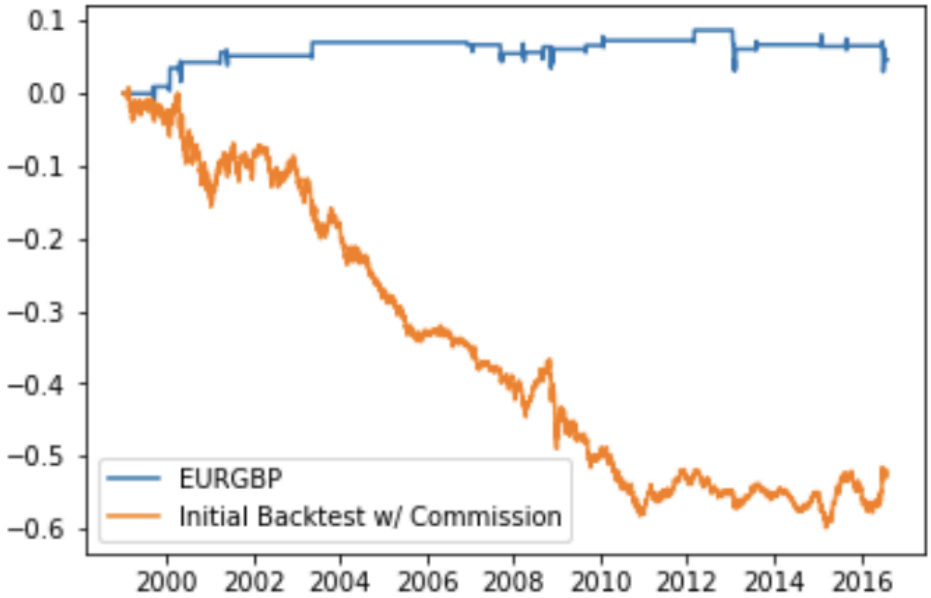

Figure 13. Optimized Backtest for EURGBP Training Data w/ Commission

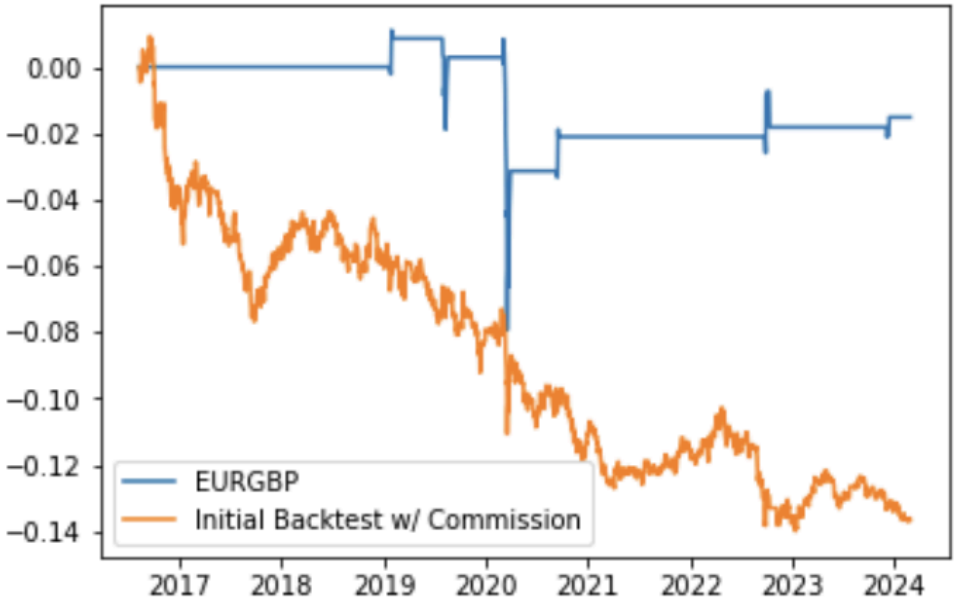

Figure 14. Optimized Backtest for EURGBP Testing Data w/ Commission

	EURGBP Train	EURGBP Test
Strategy Return (%)	4.69	-1.52
Buy and Hold Return (%)	17.68	0.82
Sharpe Ratio	0.11	0
# Trades	20	10
Avg. Trade Duration	12 days	7 days
Expectancy	0.24	-0.13

Figure 15. Optimized EURGBP w/ Commission Backtest Metrics

It is evident that the commission case did not perform as well out of sample as the non-commission case did, however the improvement from the initial backtesting strategy is vast. Note that due to the high entry barrier, the strategy only made 20 trades over the course of the training data and 10 trades over the course of the test data.

4. Robustness

After outlining the strategy process, we now repeat section 3 for every currency to test the strategies effectiveness on other spaces within the FX market. But firstly, it is important to conduct some data analysis to come up with hypotheses for how well our strategy will perform.

4.1 Mean Reversion Statistics

Given that the Bollinger Bands strategy is one based on mean reversion, it is important to have statistics to measure the level of mean reversion of our data. I used the Augmented Dickey-Fuller (ADF) Test to evaluate the data. The ADF test is a statistical test for stationarity, specifically testing for the presence of a unit root in a time series. A negative and statistically significant test statistic suggests that there is evidence to reject the unit root null hypothesis and suggests mean reverting behavior of the time series.

4.2 Currency Data Analysis

Currency	Daily Data ADF Test Statistic	Currency	Hourly Data ADF Test Statistic
EUR-CAD	-3.38	EUR-GBP	-3.04
HKD	-3.30	EUR-HUF	-2.60
NZD-USD	-2.54	NZD-USD	-2.52
SGD	-2.45	INR	-2.48
THB	-2.18	JPY	-2.46

Figure 16. Most Significant ADF Test Statistics for Each Data Type

Above is displayed the most significant ADF test statistics for the daily and hourly currency time series, ordered by the ADF test statistic. Using this table along with the full table in the appendix we have an idea of which markets we should expect the Bollinger Bands strategy will perform best in.

1. There is no dataset in which the ADF test statistic is extremely significant. This suggests that the strategy might not work as well on just a raw currency pair, and better results could be obtained by creating a hedged portfolio as done in the initial example.
2. We should expect our strategy to perform best in daily EUR-CAD, daily HKD, and hourly EUR-GBP.
3. We should expect better performance over all currencies in the hourly data compared to the daily data.

4.3 Full Data Performance

First we analyze the chosen optimal parameters from just using the training dataset. The table below shows the average parameter for each dataset.

Dataset/Parameter Average	Lookback Period	EntryZ	ExitZ
Daily	42.8	20	5.2
Daily with Commission	90.5	31.9	6.9
Hourly	101	26.4	4.3
Hourly with Commission	122.6	33.1	6.4

Figure 17. Average Optimized Parameters Across Each Data Type

We see that daily data with no commission had a shortest lookback period. This is not surprising from earlier results, since shorter lookback periods lead to more trades with smaller profits on each trade, and a no commission environment makes those more profitable. Also unsurprisingly, commission environments led to higher entryZscores. On average exitZscores were also higher for commission environments, but not by too much.

	Return (%)	Buy and Hold Return (%)	# Trades	Trade Duration	Sharpe Ratio	Expectancy (%)
Daily (Train)	101.58	11.02	247.38	15.43 days	0.24	0.28
Daily with Commission (Train)	16.99	11.02	25.67	28.57 days	0.20	1.13
Hourly (Train)	984.29	4.17	199.57	1.55 hours	0.90	0.10
Hourly with Commission (Train)	56.57	4.17	87	2.2 hours	0.46	0.07

Figure 18. Average Optimized Backtest Metrics Across Each Training Data Type

	Return (%)	Buy and Hold Return (%)	# Trades	Average Trade Duration	Sharpe Ratio	Expectancy (%)
Daily	4.53	11.88	96.38	14.76 days	0.22	0.11
Daily with Commission	3.93	11.88	9.33	28 days	0.24	1.17
Hourly	6.69	1.08	81.48	1.5 hours	1.06	0.09
Hourly with Commission	-2.36	1.08	34.83	2.9 hours	0.15	-0.07

Figure 19. Average Optimized Backtest Metrics Across Each Testing Data Type

It is clear that on average, trade durations for hourly data are far shorter than daily data. Additionally, trade durations for commission environments seem to be double that of non-commission environments. Hourly no commission data performs the best out of sample. While this is carried by an outlier performance of the strategy on the Russian Ruble, the out of

sample testing is still decent without it. It is also important to note the high average sharpe ratio for the hourly no commission data, both in and out of sample.

When analyzing the full, non-averaged results, it is clear that certain datasets are simply more potent than others for mean reversion strategies. Most of the top ranked strategies in the previous data analysis section beat out the buy and hold and generate positive returns in all datasets except for hourly commission. Additionally, highly volatile datasets such as RUB perform very well in hourly data testing. Ultimately, simple currency markets outside of the highest ranked ones for mean reversion will not perform well under this strategy. Full table results are shown in the appendix.

[More detailed full analysis coming soon]

5. Appendix

	Daily Data ADF Test-Statistics	Hourly Data ADF Test-Statistics
EURGBP	-1.553344	-3.038054
EURHUF	-0.644675	-2.599025
NZDUSD	-2.538556	-2.520047
INR	0.061989	-2.476632
JPY	-1.249234	-2.456117
HKD	-3.303217	-2.439036
AUDUSD	-1.870751	-2.419092
PHP	-1.232977	-2.364250
THB	-2.180903	-2.318849
CNY	-1.821421	-2.288438
EURUSD	-1.863414	-2.205406
EURCHF	-0.229041	-1.912158
RUB	-0.809824	-1.848319
ZAR	-0.286893	-1.791804
SGD	-2.454717	-1.650999
MYR	-0.465669	-1.639869
EURSEK	-1.319795	-1.466242
EURJPY	-1.707377	-1.374862
MXN	-1.389508	-1.206081
EURCAD	-3.382603	-1.143155
GBPJPY	-1.427016	-1.025845

Figure 5.1 Full ADF Test Statistics for All Data

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Avg. Trade Duration	Expectancy [%]
EURCAD train	39.58	-1.66	0.2818	215	42.791	16	0.17285
HKD train	0.83	0.08	0.0727	25	56.000	32	0.03338
NZDUSD train	28.13	14.82	0.2158	52	61.538	26	0.54764
SGD train	1.67	-23.91	0.1109	2	50.000	56	0.85303
THB train	41.70	-21.53	0.3701	428	50.000	3	0.08682
AUDUSD train	924.24	-7.80	0.0704	453	60.265	8	1.25404
EURUSD train	7.66	4.55	0.0567	413	54.237	7	0.02763
CNY train	10.98	-19.24	0.2863	49	51.020	3	0.22062
EURJPY train	16.09	4.37	0.0894	273	50.183	14	0.08114
EURGBP train	84.02	17.68	0.3991	430	66.744	13	0.15216
GBPJPY train	5.27	-18.08	0.0898	27	59.259	11	0.21391
MXN train	51.17	63.84	0.2592	668	63.473	5	0.07108
JPY train	24.32	5.92	0.4779	12	91.667	29	1.87705
EURSEK train	48.56	1.22	0.3029	375	60.000	15	0.11283
PHP train	3.34	-7.36	0.1277	15	60.000	22	0.22461
RUB train	-3.61	93.61	0.0000	5	20.000	7	-0.72198
EURHUF train	54.53	15.43	0.3853	468	57.051	7	0.10289
MYR train	649.54	2.98	1.0140	834	64.149	6	0.25086
ZAR train	126.42	86.87	0.2628	276	68.116	17	0.35581
EURCHF train	21.38	-20.61	0.2430	165	46.667	18	0.12629
INR train	-2.59	40.30	0.0000	10	50.000	9	-0.25447

Figure 5.2 Full Daily Training Backtest Metrics

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Avg. Trade Duration	Expectancy [%]
EURCAD test	10.49	-6.25	0.2616	82	53.659	17	0.13095
HKD test	0.89	0.15	0.3599	8	62.500	39	0.11060
NZDUSD test	12.92	-16.08	0.3449	31	54.839	20	0.42252
SGD test	5.38	2.58	0.7042	2	100.000	35	2.65357
THB test	-18.02	14.56	0.0000	154	51.948	4	-0.12622
AUDUSD test	-20.68	-7.98	0.0000	194	53.093	8	-0.10966
EURUSD test	10.13	-13.44	0.3020	175	54.286	7	0.05968
CNY test	-0.31	7.09	0.0000	25	48.000	3	-0.01184
EURJPY test	22.84	21.69	0.4524	110	49.091	15	0.19874
EURGBP test	37.14	0.83	0.4260	186	60.215	13	0.17419
GBPJPY test	3.52	23.22	0.1648	12	75.000	8	0.31567
MXN test	3.17	-7.77	0.0453	280	56.786	6	0.02421
JPY test	-0.75	24.97	0.0000	4	50.000	40	-0.06728
EURSEK test	13.73	17.35	0.2726	139	58.273	17	0.09719
PHP test	1.68	8.81	0.1296	8	50.000	14	0.21314
RUB test	-9.15	62.35	0.0000	4	50.000	5	-2.24631
EURHUF test	24.04	25.42	0.5176	200	56.000	8	0.11831
MYR test	-32.78	23.05	0.0000	240	50.833	9	-0.16172
ZAR test	20.10	58.67	0.1875	99	64.646	18	0.22498
EURCHF test	10.30	-18.69	0.4143	65	47.692	18	0.15620
INR test	0.54	29.07	0.0998	6	66.667	6	0.09035

Figure 5.3 Full Daily Testing Backtest Metrics

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Avg. Trade Duration	Expectancy [%]
EURCAD test	1.41	-6.25	0.0530	11	45.455	64	0.1373
HKD test	-0.35	0.15	0.0000	7	42.857	35	-0.0496
NZDUSD test	9.99	-16.08	0.6142	2	100.000	26	4.9068
SGD test	3.78	2.58	0.5267	3	100.000	20	1.2490
THB test	-4.04	14.56	0.0000	5	20.000	10	-0.8195
AUDUSD test	6.94	-7.98	0.3337	1	100.000	73	6.9363
EURUSD test	3.15	-13.44	0.6825	1	100.000	4	3.1495
CNY test	0.21	7.09	0.0636	5	80.000	6	0.0413
EURJPY test	16.29	21.69	0.6069	7	85.714	54	2.2050
EURGBP test	-1.52	0.83	0.0000	10	60.000	7	-0.1306
GBPJPY test	20.88	23.22	0.7035	12	83.333	20	1.6305
MXN test	1.45	-7.77	0.1624	2	50.000	8	0.7311
JPY test	-1.55	24.97	0.0000	4	50.000	40	-0.2537
EURSEK test	9.39	17.35	0.3065	27	51.852	28	0.3419
PHP test	-0.11	8.81	0.0000	1	0.000	71	-0.1064
RUB test	12.02	62.35	0.1131	4	75.000	5	2.9956
EURHUF test	14.73	25.42	0.4178	29	58.621	21	0.4988
MYR test	-17.26	23.05	0.0000	50	38.000	12	-0.3732
ZAR test	7.10	58.67	0.1776	8	62.500	25	1.0649
EURCHF test	1.06	-18.69	0.1955	2	50.000	55	0.5334
INR test	-0.94	29.07	0.0000	5	0.000	4	-0.1894

Figure 5.4 Full Daily w/ Commission Training Backtest Metrics

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Avg. Trade Duration	Expectancy [%]
EURCAD train	28.78	-1.66	0.4589	11	72.727	65	2.3805
HKD train	-3.33	0.08	0.0000	22	18.182	32	-0.1538
NZDUSD train	5.70	14.82	0.2481	2	100.000	33	2.8354
SGD train	3.46	-23.91	0.2036	4	75.000	21	0.8602
THB train	3.12	-21.53	0.2001	9	33.333	3	0.3525
AUDUSD train	105.20	-7.80	0.2188	10	60.000	74	9.5924
EURUSD train	6.02	4.55	0.3020	4	75.000	15	1.5009
CNY train	-1.33	-19.24	0.0000	16	12.500	6	-0.0826
EURJPY train	6.08	4.37	0.0534	27	51.852	38	0.3283
EURGBP train	4.70	17.68	0.1139	20	65.000	12	0.2361
GBPJPY train	30.06	-18.08	0.3210	27	55.556	28	1.0315
MXN train	1.69	63.84	0.1073	2	50.000	9	0.8486
JPY train	21.71	5.92	0.4346	12	91.667	29	1.6934
EURSEK train	26.61	1.22	0.2614	85	55.294	29	0.2936
PHP train	0.97	-7.36	0.0433	7	57.143	37	0.1394
RUB train	-7.25	93.61	0.0000	5	0.000	23	-1.4827
EURHUF train	14.20	15.43	0.1628	71	57.746	17	0.2130
MYR train	63.21	2.98	0.5354	165	59.394	7	0.3102
ZAR train	22.85	86.87	0.1206	22	54.545	30	1.0395
EURCHF train	26.88	-20.61	0.3212	12	58.333	87	2.2005
INR train	-2.53	40.30	0.0000	6	33.333	5	-0.4200

Figure 5.5 Full Daily w/ Commission Testing Backtest Metrics

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Max. Trade Duration	Expectancy [%]
EURCAD train	11.62	4.49	3.0761	85	71.7647	3	0.12985
HKD train	-0.19	0.00	0.0000	4	25.0000	1	-0.04738
NZDUSD train	22.85	-6.47	1.4422	257	42.0233	6	0.08232
SGD train	2.71	-2.56	0.6508	56	55.3571	10	0.04882
THB train	10.74	5.92	1.2125	473	60.0423	3	0.02206
AUDUSD train	16.79	-6.02	1.0890	265	40.3774	6	0.06095
EURUSD train	10.72	0.19	0.9500	178	39.3258	9	0.05906
CNY train	0.44	14.82	0.3287	38	52.6316	2	0.01183
EURJPY train	10.29	20.34	1.1693	46	50.0000	15	0.21830
EURGBP train	10.76	2.62	1.4526	83	48.1928	18	0.12497
GBPJPY train	9.56	17.42	1.4652	18	61.1111	8	0.53192
MXN train	6.11	-18.09	0.6726	34	55.8824	12	0.17993
JPY train	2.88	20.21	0.3836	24	50.0000	7	0.12516
EURSEK train	1.95	8.37	0.3446	91	53.8462	5	0.02197
PHP train	-1.14	8.59	0.0000	20	40.0000	4	-0.05702
RUB train	20522.22	-13.29	1.1382	1744	59.6330	3	0.31928
EURHUF train	11.26	1.33	1.1321	39	51.2821	15	0.28359
MYR train	-0.85	9.42	0.0000	39	56.4103	3	-0.02147
ZAR train	10.93	17.36	0.6979	168	43.4524	10	0.06581
EURCHF train	7.23	-6.01	0.9578	399	74.9373	8	0.01777
INR train	3.31	8.88	0.6710	130	52.3077	8	0.02561

Figure 5.5 Full Hourly Training Backtest Metrics

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Avg. Trade Duration	Expectancy [%]
EURCAD test	1.82	-2.16	1.2585	35	62.8571	1	0.05184
HKD test	0.00	0.10		0			
NZDUSD test	14.27	-3.14	2.7807	101	46.5347	1	0.13370
SGD test	1.26	1.89	1.0207	24	62.5000	2	0.05249
THB test	1.22	3.88	0.3604	205	60.4878	0	0.00639
AUDUSD test	5.40	-4.15	1.2499	101	43.5644	1	0.05361
EURUSD test	2.38	-4.02	0.7451	75	42.6667	2	0.03218
CNY test	-0.04	-0.74	0.0000	23	52.1739	0	-0.00176
EURJPY test	6.00	4.09	2.5509	15	66.6667	3	0.39525
EURGBP test	1.92	-0.25	1.1294	39	41.0256	2	0.04921
GBPJPY test	3.15	4.47	1.2915	8	62.5000	3	0.39265
MXN test	3.10	1.72	0.7804	29	44.8276	2	0.10868
JPY test	1.32	8.41	0.9862	9	55.5556	1	0.14969
EURSEK test	4.61	-2.14	2.1789	43	60.4651	1	0.10552
PHP test	-0.58	0.58	0.0000	5	40.0000	1	-0.11450
RUB test	84.95	2.19	2.2141	705	57.3050	0	0.08960
EURHUF test	1.36	3.92	0.5944	15	40.0000	4	0.09116
MYR test	-0.47	4.16	0.0000	15	46.6667	1	-0.03007
ZAR test	11.32	4.60	1.7539	66	48.4848	1	0.16602
EURCHF test	-2.80	-1.88	0.0000	135	77.0370	1	-0.02067
INR test	0.32	1.26	0.2662	63	46.0317	1	0.00517

Figure 5.6 Full Hourly Testing Backtest Metrics

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Avg. Trade Duration	Expectancy [%]
EURCAD train	4.52	4.49	1.1724	13	69.231	4	0.3421
HKD train	-0.81	0.00	0.0000	4	0.000	0	-0.2025
NZDUSD train	6.48	-6.47	0.8013	41	53.659	2	0.1565
SGD train	0.57	-2.56	0.2463	12	66.667	4	0.0480
THB train	2.05	5.92	0.7503	8	87.500	3	0.2557
AUDUSD train	6.55	-6.02	0.9598	17	70.588	3	0.3778
EURUSD train	5.64	0.19	1.3828	10	80.000	3	0.5519
CNY train	-3.91	14.82	0.0000	17	47.059	2	-0.2330
EURJPY train	5.79	20.34	0.9846	17	58.824	3	0.3362
EURGBP train	-0.30	2.62	0.0000	42	45.238	4	-0.0054
GBPJPY train	7.35	17.42	1.1232	13	61.538	5	0.5759
MXN train	1.38	-18.09	0.1784	23	43.478	3	0.0644
JPY train	-1.39	20.21	0.0000	23	52.174	2	-0.0532
EURSEK train	-1.46	8.37	0.0000	15	46.667	2	-0.0978
PHP train	-0.55	8.59	0.0000	3	33.333	0	-0.1843
RUB train	1160.23	-13.29	1.0545	1521	50.953	0	0.1822
EURHUF train	3.24	1.33	0.9858	15	66.667	1	0.2185
MYR train	-6.72	9.42	0.0000	17	29.412	2	-0.4057
ZAR train	-0.40	17.36	0.0000	2	0.000	0	-0.2013
EURCHF train	-0.13	-6.01	0.0000	13	46.154	3	-0.0086
INR train	-0.19	8.88	0.0000	1	0.000	0	-0.1914

Figure 5.7 Full Hourly w/ Commission Training Backtest Metrics

	Return [%]	Buy & Hold Return [%]	Sharpe Ratio	# Trades	Win Rate [%]	Avg. Trade Duration	Expectancy [%]
EURCAD test	-2.32	-2.16	0.0000	7	28.571	5	-0.3330
HKD test	0.00	0.10		0			
NZDUSD test	1.09	-3.14	0.3339	23	56.522	3	0.0495
SGD test	-0.92	1.89	0.0000	7	28.571	3	-0.1310
THB test	-0.80	3.88	0.0000	6	33.333	3	-0.1337
AUDUSD test	0.07	-4.15	0.0349	8	37.500	2	0.0103
EURUSD test	-1.67	-4.02	0.0000	7	42.857	8	-0.2388
CNY test	-1.96	-0.74	0.0000	8	25.000	2	-0.2466
EURJPY test	0.69	4.09	0.3588	3	66.667	8	0.2350
EURGBP test	-0.66	-0.25	0.0000	16	50.000	4	-0.0412
GBPJPY test	1.58	4.47	0.6428	7	57.143	3	0.2311
MXN test	-2.30	1.72	0.0000	12	25.000	3	-0.1898
JPY test	0.37	8.41	0.2486	6	66.667	2	0.0672
EURSEK test	-2.01	-2.14	0.0000	8	37.500	2	-0.2522
PHP test	-0.39	0.58	0.0000	2	0.000	0	-0.1960
RUB test	-37.19	2.19	0.0000	575	40.696	0	-0.0788
EURHUF test	-0.88	3.92	0.0000	8	50.000	2	-0.1124
MYR test	-0.16	4.16	0.0000	6	33.333	2	-0.0259
ZAR test	0.28	4.60	1.2663	1	100.000	0	0.2808
EURCHF test	-2.47	-1.88	0.0000	12	41.667	4	-0.2070
INR test	0.00	1.26		0			

Figure 5.8 Full Hourly w/ Commission Testing Backtest Metrics

6. References

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